A DRONE WAR

Hundreds of drones were deployed in the conflict. Here are some Unmanned Aerial Vehicles (UAVs) that the adversaries used



INDIA

HAROP: Israeli-made loitering munition/ kamikaze drone for precision strikes

HERON: Israeli ISR (Intelligence-Surveillance-Reconnaissance) drone

SKYSTRIKER: Israeli loitering munition

NETRA: Indian-made (IdeaForge) tactical ISR UAV

SWITCH: Indian tactical ISR/ strike UAV

OP SINDOOR: Indian Harop drones destroyed a Pakistani LY80 radar in Lahore. Other UAVs conducted strikes and cued aircraft—by providing real-time intelligence and surveillance—thus limiting manned aircraft at risk

PAKISTAN

ASISGUARD SONGAR: Turkey-made 300–400 surveillance/strike drones.

BAYRAKTAR TB2: Turkish armed ISR/ strike drone

BAYRAKTAR AKINCI: Turkish advanced ISR/ strike drone

YIHA-III KAMIKAZE: Loitering munitions used in swarms, from Turkey

OP SINDOOR: Pakistan used 300-400 Songars, 400-500 YIHA-IIIs in swarms to attack India early on; later, Bayraktar & Akinci drones targeted air bases/ installations. India's AD system shot most down



EXPERTS SAY INDIA MUST NOT BE COMPLACENT DESPITE ITS STRONG SHOWING. THE IAF'S FALLING SQUADRON STRENGTH, AND THE SLOW INDUCTION OF FIFTH GENERATION JETS LEAVE GAPS THAT CAN BE EXPLOITED

The first layer comprises counter drone systems like D4 and MANPADS (Man-Portable Air Defence Systems) like Igla-M and Igla-S, while the second layer has low-level air defence (LLAD) guns like Bofors L-70 and ZSU-23-4 Schilka self-propelled system, and short-range surface-to-air missiles (SAMs) like Pechora, Tunguska and OSA-AK. The third and fourth layers have medium- and long-range SAMs like Pechora, SPYDER, the indigenous Akash and the long-range S-400 Triumf and Barak-8 missile defence systems. India's counter-unmanned aerial system (UAS) technologies also played a major role in detecting and neutralising Pakistani drones. This multi-layered air defence network not only protected Indian air bases, army installations, airports and cities, but also protected Amritsar's Golden Temple from a Pakistani drone and missile attack.

In contrast, Pakistan's Chinese HQ-9 and HQ-16 air defence systems failed to detect and intercept the devastating Indian strikes on its air bases on May 10. Early on May 10, Indian strikes took out a Chinese-made LY80 air defence system using a Harpy kamikaze drone at Lahore, while a missile destroyed an HQ-9 system in Karachi. Comparative analysis also suggests that the S-400 surpasses the HQ-9 in performance. "India's multi-layered air defence systems exhibited greater effectiveness in deployment and integration than their Pakistani counterparts," a key defence official said.

Defence analysts say that for the past two decades, as a probable counter to India's Cold Start Doctrine, which envisages short, swift strikes into Pakistan that would avoid a nuclear escalation, Pakistan has been buying mostly offensive weaponry. "In contrast, India's strategic focus was on technological advancement and indigenous production, leading to investments in radars such as the ADTCR, Ashwini and Indra (electronically scanned array radars), sensor processing systems, jamming devices, and electronic warfare systems, including drone jammers as well as counter-battery radars," says Deshmukh. The development of missile systems such as BrahMos and Akash, coupled with the timely procurement of the S-400, has provided India with a significant advantage in non-contact warfare, he adds. Thus, Pakistan's investment in offensive weapons lays bare its weak air defence capabilities, leaving its command and control exposed.

Ultimately, the conflict reaffirmed that modern warfare is driven by air power, not merely in terms of aircraft but through sensors, missiles, data networks, electronic warfare and unmanned systems. "Operation Sindoor is a testimony to the power of a capable air force," notes a senior defence analyst. "But no country can win future wars without continuous investment in modern air platforms and integrated defence networks." Despite its strong showing, experts argue that India must not become complacent. The IAF's thinning squadron strength and the slow induction of next-gen air dominance fighters leave gaps that adversaries may exploit in the future. ■